

Human-in-the-Loop Capacitive Microphone Sensors-Based Muscle Sensing System for Predictive and Adaptive Exoskeleton Assistance

Xiao Jin , Ge Gao , Wendong Wang , Ravi Vaidyanathan , Peter Childs , and Zhenhua Yu 

Abstract—Mobility impairments among older adults and individuals with neuromuscular weakness motivate the need for timely and adaptive exoskeleton assistance. This paper presents a human-in-the-loop muscle sensing and control system based on capacitive microphone sensors (CMS) that capture subtle mechanical muscle vibrations preceding observable motion. CMS signals were shown to occur 20–30 ms earlier than IMU-based kinematics, enabling anticipatory intent detection and feedforward assistive control. A five-sensor CMS array positioned over major thigh muscles is combined with a two-stage control strategy that integrates threshold-based pre-assist triggering and machine-learning-based torque refinement. Experiments across walking, stair ascent, sitting, and standing achieved over 90% classification accuracy under both non-fatigued and fatigued conditions with low latency. Robustness evaluations demonstrate stable CMS performance under realistic wearable perturbations, including perspiration and attachment variation. Extended experimental sessions (1–2 h) and preliminary feedback from five participants indicate comfortable wear and natural interaction. These results highlight the potential of CMS-based anticipatory sensing for practical wearable exoskeleton deployment in daily scenarios.

Index Terms—Human-robot interaction (HRI), muscle sensing, prosthetic limb control, capacitive microphone sensors (CMS).

Received 18 October 2025; accepted 3 February 2026. Date of publication 19 February 2026; date of current version 2 March 2026. This article was recommended for publication by Associate Editor S. Ding and Editor H. Yu upon evaluation of the reviewers' comments. This work was supported by the Capital Equipment funding of University of Aberdeen under Grant CF10834-42. (Corresponding author: Zhenhua Yu.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Physical Sciences and Engineering Ethics Board of University of Aberdeen under Application No. 8891100.

Xiao Jin and Peter Childs are with the Dyson School of Design Engineering, Imperial College London, SW7 2AZ London, U.K.

Ge Gao is with the Department of Computer Science, University of Aberdeen, AB24 3UE Aberdeen, U.K.

Wendong Wang is with the School of Mechanical Engineering, Northwestern Polytechnical University, Xi'an 710072, China.

Ravi Vaidyanathan is with the Department of Mechanical Engineering, Imperial College London, SW7 2AZ London, U.K.

Zhenhua Yu is with the Dyson School of Design Engineering, Imperial College London, SW7 2AZ London, U.K., also with the Department of Computer Science, University of Aberdeen, AB24 3UE Aberdeen, U.K., and also with the Department of Mechanical Engineering, Imperial College London, SW7 2AZ London, U.K. (e-mail: zhenhua.yu@abdn.ac.uk).

This article has supplementary downloadable material available at <https://doi.org/10.1109/LRA.2026.3666391>, provided by the authors.

Digital Object Identifier 10.1109/LRA.2026.3666391

I. INTRODUCTION

WITH the rapid acceleration of global population aging, mobility impairment has become a major public health concern. The population aged 60 and above is projected to increase from 1 billion in 2020 to 2.1 billion by 2050, accounting for 22% of the global population [1], [2]. Mobility limitations—such as difficulty walking, stair climbing, or rising from a chair—are highly prevalent among older adults, affecting approximately one-third of individuals aged 70 and most adults over 80 [3], and are closely associated with increased risk of injury, hospitalization, and loss of independence. To address this growing demand, substantial efforts have been devoted to robotic exoskeletons for mobility assistance and rehabilitation [4], [5]. These systems aim to augment muscle function, reduce joint loading, and support daily activities, with representative examples including metabolic-cost-reducing ankle exoskeletons [6], load-bearing lower-limb systems [7], and lightweight gait-assist devices for elderly users.

Recent research explores diverse actuation schemes and human-centered control strategies, such as pneumatic or cable-driven actuation, passive elasticity, adaptive impedance control, EMG-based models, and learning-based approaches [8]. However, many systems remain constrained by high energy consumption, excessive weight, and reliance on predefined trajectories or manual tuning, limiting adaptability in real-world use [9]. Consequently, intention-driven control with low-profile muscle sensing has emerged as a key direction toward more responsive and personalized wearable assistance [10].

Surface electromyography (sEMG) is the most widely used modality for muscle activity sensing in exoskeleton control, with prior studies demonstrating real-time multi-joint control, adaptive upper-limb assistance, and stroke rehabilitation [11], [12], [13], [14]. More recent work has extended sEMG to dynamic tasks with unknown loads and hybrid EEG–EMG interfaces using time–frequency features and adaptive models [15], [16], [17]. However, sEMG remains sensitive to noise, electrode placement, and skin conditions, limiting its robustness in long-term, real-world use and motivating interest in alternative modalities such as mechanomyography (MMG) and MEMS-based sensing [18].

Compared with EMG, MMG is inherently robust to sweat, skin impedance variations, and electromagnetic interference, making it attractive for wearable applications. Recent studies

have demonstrated MMG feasibility in lower-limb and human-robot interaction tasks, including torque modeling during leg-lift motions [19], knee joint torque estimation in prosthesis users [20], and real-time tremor detection [21]. Nevertheless, most MMG systems remain laboratory-bound and rely on bulky sensors or rigid attachments, motivating the development of compact, wearable MMG-based sensing and control frameworks for real-world legged assistance.

When implemented using MEMS-based sensing elements, MMG systems offer advantages in sensor miniaturization, low power consumption, and ease of integration into wearable form factors, enabling sensitive capture of low-frequency mechanical vibrations associated with early muscle activation in a compact, low-profile design suitable for long-term wear. To address the remaining limitations of conventional MMG implementations in long-term wearability, sensor integration, and effective information throughput, this study proposes an assistive exoskeleton control strategy based on capacitive microphone sensors (CMS). Unlike traditional MMG approaches that focus primarily on post-contraction vibration analysis, the proposed CMS method exploits the high mechanical sensitivity of capacitive MEMS microphones to detect anticipatory muscle vibrations that precede observable limb motion, enabling early user intent recognition and a feedforward control strategy with improved temporal alignment between human motion and assistive torque delivery.

Compared with conventional EMG- or accelerometer-based systems, CMS-based sensing provides a compact, low-cost, and skin-condition-insensitive solution suitable for daily wearable use. Experimental results demonstrate that the proposed CMS-based system maintains stable recognition accuracy and output behavior under realistic perturbations, including muscle fatigue, motion variability, and attachment uncertainty, highlighting its robustness for real-world deployment. Furthermore, dynamic torque modulation allows the assistive output to adapt in real time to changes in muscle activation and physical exertion, reducing user effort while preserving natural and intuitive human-exoskeleton interaction.

The remainder of the paper is organized as follows. Section II introduces the hardware architecture, including the design of the capacitive microphone sensors, their placement on the thigh, and the assistive module structure. Section III presents the muscle activity recognition and control framework, highlighting early-onset detection, fatigue analysis, and machine learning models. Section IV describes the experimental validation under both controlled and real-world conditions, including robustness evaluation.

II. HARDWARE ARCHITECTURE

A. Capacitive Microphone Sensors Design

This design is an optimized version of a previously proposed CMS (Capacitive Microphone Sensor) architecture [22], employing a Knowles SPU0410LR5H-QB MEMS microphone (38 dBV/Pa, 20 Hz–20 kHz; $3.35 \times 2.5 \times 1.0$ mm) for its high mechanical sensitivity and compact size as shown in Fig. 1(a). The central silicone filling and bottom conical horn are retained to improve mechanical stability, suppress high-frequency noise,

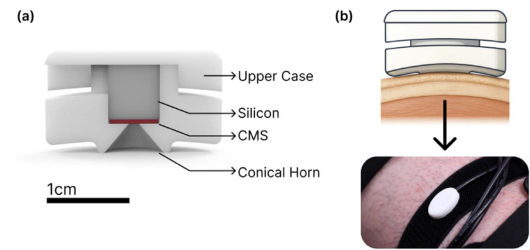


Fig. 1. (a) CMS structure, where the bottom conical cavity and silicone filling are retained from previous work to enhance low-frequency vibration concentration and suppress noise. (b) Redesigned curved skin interface that increases thigh contact area, improving attachment stability and wearing comfort during dynamic exoskeleton operation.

and enhance low-frequency vibration concentration, enabling reliable capture of sub-100 Hz muscle vibrations when coupled to skin. To improve robustness and comfort during long-term wear and dynamic movements, the skin-contact interface was redesigned with a curved surface to increase contact area, while the ring-shaped protrusion at the bottom is preserved to maintain consistent sensitivity and signal quality. Prior to integration, the MEMS microphone was calibrated following the procedure in [22] using a 1000-Hz reference excitation. All CMS components were permanently bonded to eliminate internal air gaps and ensure stable mechanical coupling. The silicone filling acts as a viscoelastic mechanical filter that suppresses high-frequency noise while reinforcing low-frequency MMG-relevant vibrations. The same silicone configuration was used throughout all experiments.

To mitigate sensor slippage during dynamic movements, the CMS module was secured using a non-elastic wearable band with an anti-slip inner lining. This configuration maintains stable contact pressure between the sensor and skin while preventing relative motion during repeated muscle contractions.

To improve reliability, flexible cables are integrated into a structured wearable band to minimize disconnection and motion-induced noise. Signals are routed along the exoskeleton frame to a compact processor unit mounted at the lower back. A Raspberry Pi 4 Model B performs real-time data acquisition, feature extraction, and logistic-regression-based torque control, enabling lightweight (45 g), fully portable operation without external computing as shown in Fig. 1(b).

B. CMS Integration on the Thigh

To enable accurate detection of muscle activity and motion intent on the thigh, five Capacitive Microphone Sensors (CMS) are strategically placed over key muscle groups as shown in Fig. 2(a): Rectus femoris, Vastus muscles, Sartorius, Medial hip adductor muscles, and Posterior knee flexor muscles. These muscles together cover the full spectrum of lower-limb functions, including hip and knee flexion/extension, adduction, and rotational control. The anterior muscles (Rectus femoris and Vastus) are essential for knee extension and propulsion during walking and standing, while the Sartorius captures complex transitions involving multi-joint coordination. The medial adductors contribute to pelvic stability and side-to-side movement, and

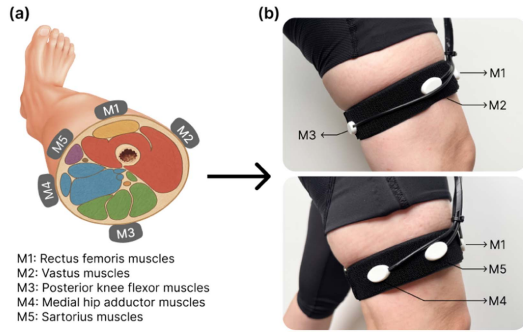


Fig. 2. Sensor placement and corresponding muscle groups. (a) Cross-sectional anatomy of the thigh showing the five targeted muscle groups: M1—Rectus femoris, M2—Vastus muscles, M3—Posterior knee flexors, M4—Medial hip adductors, and M5—Sartorius. (b) Practical sensor placement on the thigh, with each capacitive microphone sensor (white) positioned over its respective muscle group.

the posterior flexors (hamstrings) are vital for deceleration and push-off. This configuration ensures full functional coverage of the thigh and enables robust sensing of mechanical muscle activity, providing a reliable basis for responsive control in wearable assistive or rehabilitative systems as shown in Fig. 2(b).

C. Torque Estimation for Partial Assistance During Movement

This module reduces muscular effort in common lower-limb tasks such as walking, stair ascent, sitting, and standing. The sit-to-stand transition imposes the greatest hip joint demand, especially for older adults or those with weakness, making it a key target for assistance.

Rather than fully actuating the thigh, the system targets partial hip torque compensation. Using a simplified biomechanical estimate (effective mass $m = 35$ kg, $g = 9.81$ m/s², thigh length $L = 0.45$ m), the gravitational hip torque during the most demanding phase of sit-to-stand is $\tau_{\text{gravity}} \approx 77.3$ Nm. A 20% partial-compensation target at this peak-demand phase therefore defines a maximum assistive torque of $\tau_{\text{assist,max}} \approx 15.5$ Nm, which serves as the system's design upper bound for actuator sizing and safety rather than a continuously applied torque.

Assistance is delivered through a time-varying torque profile. Although the actuator is sized for a maximum of 15.5 Nm, the sustained assistive torque is set to 6 Nm under normal conditions and adaptively increased to 9 Nm when fatigue is detected, consistent with prior lower-limb exoskeleton studies showing that moderate torques effectively reduce muscular load without full joint compensation [23], [24]. To compensate for recognition and classification latency, a 3 Nm pre-assist torque is applied immediately upon early muscle activation detection, followed by a smooth transition to the sustained assistance level once motion and fatigue states are identified.

D. Design of the Prosthetic Assistance Module

The mechanical structure of the prosthetic assistance module adopts a compact and robust configuration to ensure stable torque output and user comfort during daily activities. As shown in Fig. 3, a 12 V 15 RPM 3246WG385 worm gear DC motor is employed, capable of delivering a maximum torque of 80 kgf·cm under rated conditions. The torque and rotational speed can be



Fig. 3. The prosthetic assistance module features a compact and modular drive unit composed of a 12 V worm gear DC motor and dual spur gears, rigidly linked to the thigh and anchored near the hip via an elastic strap.

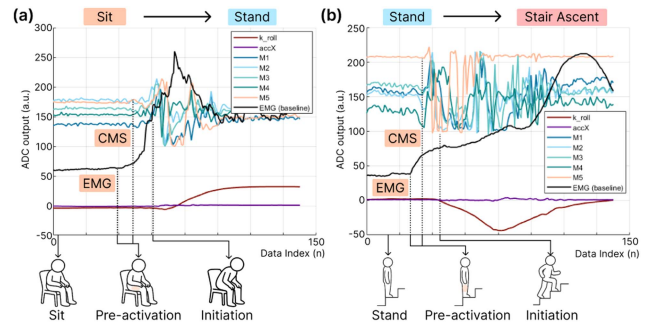


Fig. 4. Comparison of CMS and IMU signals during (a) sit-to-stand and (b) stair ascent. CMS channels (M1–M5) show earlier responses than IMU roll and acceleration and the baseline of EMG reading, demonstrating anticipatory muscle activation detection.

adjusted in real time by modulating the input current and PWM duty cycle, allowing the system to adapt to various movement demands and user fatigue levels.

The power transmission mechanism consists of two spur gears (module 1.5, 30 teeth, 15 mm face width), which are sufficient to transmit the required torque under all expected operating conditions. The outer gear is directly connected to the thigh via a rigid linkage, while the opposite end is secured near the hip joint using an elastic strap. This design ensures both mechanical efficiency and ergonomic conformity. All control electronics and the power supply are mounted on the rear side of the user's body. Signal and power cables are routed through a waist-mounted pouch to minimize movement interference and enhance system integration for wearable use.

III. MUSCLE ACTIVITY RECOGNITION AND ASSISTIVE CONTROL

Approved by the University of Aberdeen Ethics Committee (ID: 8891100), experiments involved repeated daily lower-limb tasks in a controlled lab. All channels were sampled at 1270 Hz, with data collected over two weeks to minimize fatigue effects.

A. Early-Onset Signal Recognition of Muscle Contraction via CMS

As shown in Fig. 4(a) and (b), CMS signals exhibit a consistent early-onset response compared to IMU measurements during muscle contraction. Using an integrated IMU for comparison at

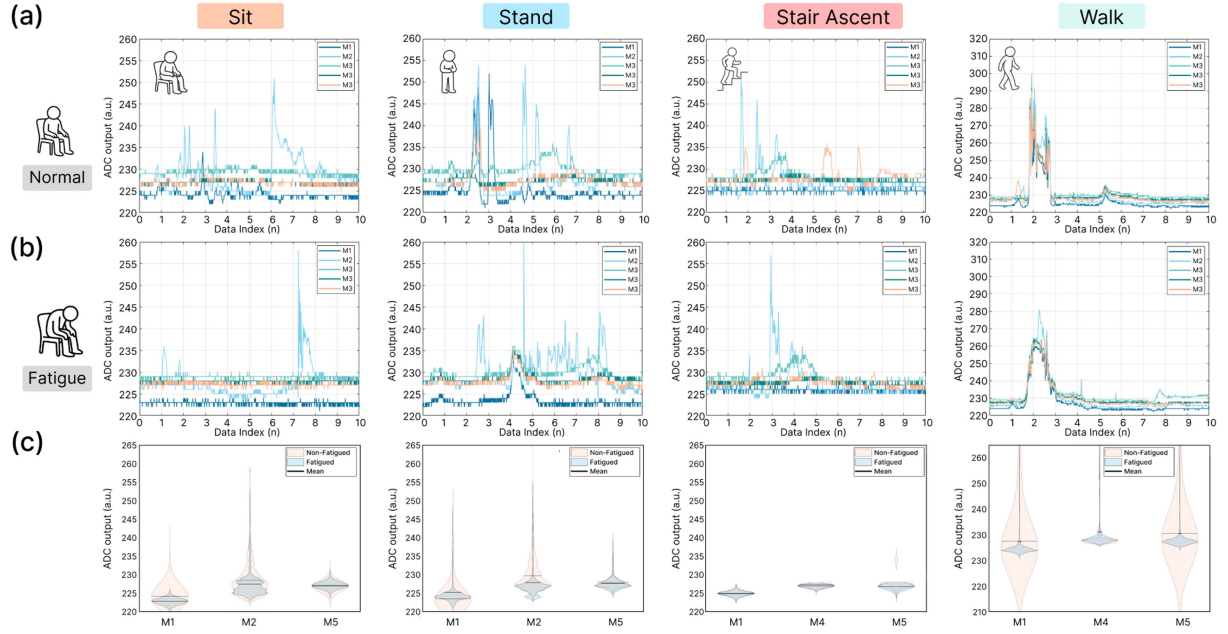


Fig. 5. Muscle vibration signals recorded by Capacitive Microphone Sensors (CMS) under different fatigue conditions. (a) Raw CMS signals during Sit, Stand, Stair Ascent, and Walk actions under normal(non-fatigued) conditions. (b) Corresponding signals under fatigued conditions, showing increased high-frequency, low-amplitude fluctuations, particularly in M1 and M2. (c) Violin plots comparing the distribution of signal amplitudes across muscle sites between non-fatigued (blue) and fatigued (orange) states, highlighting differences in signal variance and mean values, indicative of muscle fatigue effects.

340 Hz, two tasks were evaluated: sit-to-stand and stair ascent. CMS responses preceded kinematic roll changes by 26 ± 5 ms and 12 ± 3 ms, respectively. Baseline EMG signals were additionally incorporated as a physiological reference, revealing that EMG onset preceded CMS responses by approximately 10–40 ms, consistent with electromechanical delay. While EMG provides the earliest activation timing, CMS signals still emerge sufficiently early and consistently prior to observable joint motion, enabling reliable anticipatory actuation with improved robustness for long-term wearable use.

B. Muscle Activity Analysis Across Normal Movements

To investigate the neuromechanical response of lower-limb musculature during common daily activities, five major muscle groups (M1–M5) were monitored using the proposed multi-sensor array while participants performed four representative tasks: sitting down, standing up, stair ascent, and walking. All measurements were recorded under non-fatigued conditions to establish a reference profile of natural muscle activation (Fig. 5(a)).

Across tasks, CMS captured clear task-dependent and anticipatory activation patterns. Load-bearing transitions (sit-down and stand-up) were dominated by M1–M2 activity, with anticipatory M2 engagement during eccentric braking and strong M1–M2 bursts at seat-off. Stair ascent exhibited immediate M2 activation for knee extension, followed by coordinated M5 and M4 contributions during leg lifting and hip control, while walking showed compact, rhythmic activation across all muscles, initiated by M5 during leg lift.

Overall, M2 consistently contributed to vertical force generation, whereas M5 played a key role in dynamic limb motion.

These structured, task-specific activation patterns provide discriminative features for robust action recognition and subsequent fatigue analysis.

C. Muscle Activity Under Fatigued Conditions

To examine the effects of muscular fatigue, a second measurement session was conducted following an extended repetition protocol inducing localized lower-limb fatigue, during which the same four functional tasks were repeated for direct comparison with baseline conditions (Fig. 5(b)).

Under fatigue, CMS revealed systematic changes in muscle activation patterns across tasks. Load-bearing transitions were characterized by reduced overall amplitude and altered recruitment, with dominant contributions shifting toward M2 and diminished involvement of M1. Stand-up movements exhibited slower dynamics and low-amplitude, high-frequency fluctuations, indicative of unstable contraction and compensatory control. During stair ascent, M2 and M4 remained dominant, while dynamic M5 responses were markedly attenuated. Walking showed delayed onset, reduced peak amplitudes, and prolonged decay despite comparable speed (Fig. 5(c)).

Overall, muscular fatigue resulted in diminished activation magnitude, slower response dynamics, and reduced contraction efficiency, particularly in power-intensive tasks. These fatigue-induced changes underscore the importance of real-time, muscle-state-aware assistive control.

D. Cross-Subject Consistency Across Representative Movements

Five able-bodied participants (three males and two females) were recruited for the cross-subject consistency analysis as

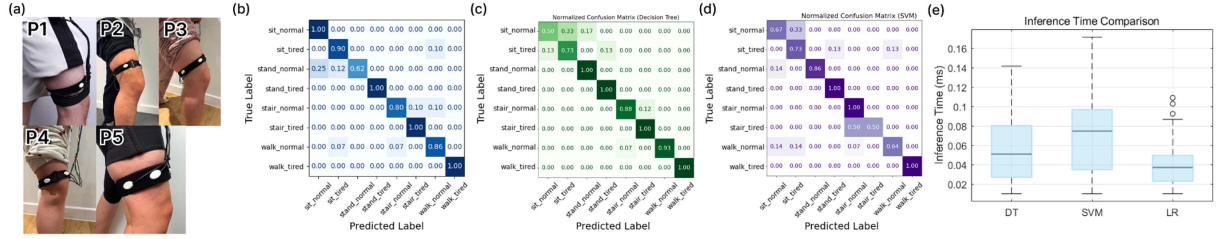


Fig. 6. (a) CMS sensor placement on the thigh across five able-bodied participants (P1–P5) with varying physical characteristics, illustrating consistent wearable deployment. (b)–(d) Normalized confusion matrices for Logistic Regression (84.9%), Decision Tree (88%), and SVM (81%) using 600-sample (~ 0.47 s) windows. (e) Inference time comparison among DT, SVM, and LR models.

TABLE I

PHYSICAL CHARACTERISTICS OF PARTICIPANTS INVOLVED IN THE CROSS-SUBJECT CONSISTENCY ANALYSIS

ID	Age	Sex	Weight(kg)	Height(cm)	Body type
P1	24	M	76	175	Average
P2	28	M	80	168	Higher adiposity
P3	24	M	85	176	Solid
P4	23	F	73	167	Solid
P5	28	F	66	157	Lean

TABLE II

CMS CHANNEL ABLATION ANALYSIS BASED ON STRATIFIED FIVE-FOLD CROSS-VALIDATION

Configuration	Accuracy (%)	Δ (%)
All 5 channels	81.55 ± 1.97	—
w/o CMS1	72.74 ± 4.10	−8.81
w/o CMS2	76.31 ± 4.90	−5.25
w/o CMS3	79.35 ± 2.83	−2.20
w/o CMS4	81.54 ± 1.83	−0.01
w/o CMS5	81.00 ± 2.42	−0.55

shown in Fig. 6(a). Participants exhibited heterogeneous physical characteristics, with body mass ranging from 66–85 kg and height from 157–176 cm, covering a range of body compositions from relatively lean to higher adiposity as well as varying levels of muscularity in Table I. All participants reported no known neuromuscular, musculoskeletal, or mobility impairments.

Each participant performed the same set of representative daily movements—sitting down, standing up, stair climbing, and walking—with repeated trials under identical conditions, using a consistent CMS placement protocol to ensure comparable sensor–skin coupling. Signals from five CMS channels were amplitude-normalized, fused using principal component analysis (PCA), and summarized by the root-mean-square (RMS) of the first principal component as a compact movement descriptor. Cross-subject consistency was quantified using a two-way random-effects, single-measurement intraclass correlation coefficient [ICC(2,1)], yielding values of 0.95 (sitting), 0.89 (standing), 0.84 (stair climbing), and 0.80 (walking). These results indicate good-to-excellent agreement across movements, with higher consistency for constrained postural transitions and slightly reduced, yet robust, agreement for more dynamic locomotor tasks.

E. Feature Extraction and Machine Learning Models

To quantify differences in muscle responses between normal and fatigued conditions, raw signals from five MEMS-based muscle sensors were processed into fixed-length feature vectors. Each trial consisted of 200 to 1000 samples per channel, and from each channel, four time-domain features—*mean*, *standard deviation*, *maximum*, and *minimum*—were extracted, resulting in a 20-dimensional input vector for classification ($\mathbf{x} \in \mathbb{R}^{20}$). Given the real-time constraints of assistive exoskeleton systems, multinomial Logistic Regression (LR) was adopted as the core classifier due to its low latency, high interpretability, and

minimal computational load. For each class k , a linear activation score is computed and passed through a Softmax function to yield the class probability:

$$P(y = k | \mathbf{x}) = \frac{e^{\mathbf{w}_k^T \mathbf{x} + b_k}}{\sum_j e^{\mathbf{w}_j^T \mathbf{x} + b_j}} \quad (1)$$

The model is trained by minimizing the cross-entropy loss over all N training samples, and the final prediction is given by the most probable class:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log P(y^{(i)} | \mathbf{x}^{(i)}), \quad \hat{y} = \arg \max_k P(y = k | \mathbf{x}) \quad (2)$$

As shown in Fig. 6(b), At 600 samples (~ 0.47 s), classification accuracy exceeded 87% while maintaining sub-second response time, providing a favorable trade-off between performance and latency; this window was therefore selected for subsequent evaluation. Using stratified five-fold cross-validation at the trial level, the Logistic Regression classifier achieved $87.1\% \pm 2.0\%$ accuracy. A stratified 80/20 train–test split (random_state = 42) yielded a test accuracy of 84.9%, consistent with the cross-validation results. The mean per-sample inference latency was 0.043 ± 0.003 ms, confirming suitability for real-time exoskeleton control.

To assess the contribution of individual CMS channels, we conducted leave-one-out sensor ablation analyses using a stratified cross-validation protocol as shown in Table II. Removing CMS1–CMS3 led to a clear reduction in classification accuracy, whereas removing CMS4 or CMS5 resulted in only marginal performance changes. In contrast, single-channel classifiers achieved substantially lower accuracy than the full five-channel configuration. Overall, these results indicate that the CMS channels provide complementary muscle vibration information and

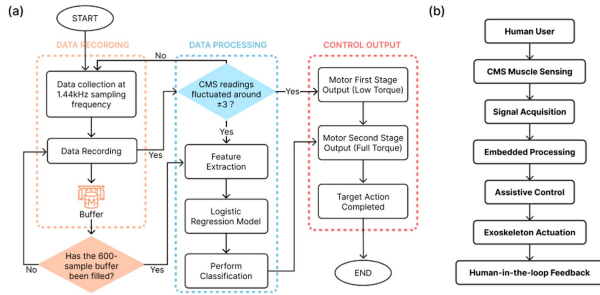


Fig. 7. (a) Real-time CMS-based control logic with two-stage assistive torque. (b) High-level closed-loop overview from muscle sensing to exoskeleton actuation.

support the rationale for the proposed multi-channel configuration.

To benchmark classification performance beyond logistic regression, Decision Tree (DT) and Support Vector Machine (SVM) classifiers were evaluated using identical 600-sample windows and test sets. As shown in Fig. 6(c) and (d), although DT achieved the highest overall accuracy (88%), its performance on the critical sit_normal class was poor (50%), limiting reliability for daily postural recognition. SVM reached 81% accuracy but showed higher inference-time variance and reduced robustness on key classes. In contrast, logistic regression achieved competitive accuracy with balanced performance across all eight classes while exhibiting the lowest inference latency (0.033 ms), compared with DT (0.054 ms) and SVM (0.068 ms) (Fig. 6(e)).

Given its low latency, stable performance, and ease of deployment, logistic regression with 600-sample windows was selected as the final classifier. At a sampling rate of 1270 Hz, this window corresponds to approximately 0.47 s, remaining within the early phase of lower-limb motion and enabling timely torque adaptation without sacrificing recognition accuracy.

IV. EXPERIMENTAL VALIDATION

A. Human-in-The-Loop Design for Exoskeleton Activation

As shown in Fig. 7, signals from five muscle groups are continuously sampled and processed using a sliding-window scheme. Muscle contraction is first detected via threshold-based triggering to initiate assistance with a low initial torque (3 Nm) for movement onset. Subsequently, CMS features are extracted and classified using logistic regression to estimate the user's action and fatigue state, which guides a second-stage adaptive torque adjustment. This two-stage strategy enables both prompt assistance initiation and smooth, personalized support.

The total system latency in this human-in-the-loop control framework is composed of three components:

$$T_{\text{total}} = T_{\text{sensor}} + T_{\text{classification}} + T_{\text{control}} \quad (3)$$

where T_{sensor} is the window buffering time (e.g., 0.42 s for 600 samples), $T_{\text{classification}}$ is the inference time of the logistic regression model, and T_{control} accounts for signal transmission and motor actuation delay.

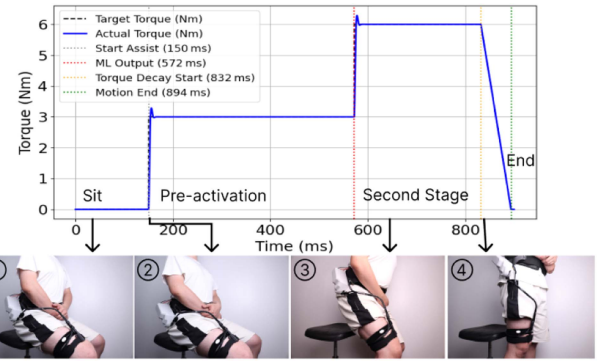


Fig. 8. Real-world sit-to-stand assistance using CMS-based control. Early muscle vibration detection enables pre-activation (3 Nm) before visible motion, followed by ML-triggered full assistance (6 Nm) and smooth torque decay to motion completion.

TABLE III
RECOGNITION ACCURACY (%) FOR EACH MOVEMENT UNDER FATIGUE AND NON-FATIGUE CONDITIONS

Movement Type	Non-Fatigued Accuracy	Fatigued Accuracy
Walking	90%	100%
Ascending Stairs	80%	90%
Sit Down	90%	90%
Stand Up	80%	80%

B. Exoskeleton System Performance in Real World

To assess real-world applicability, CMS-based motion recognition and assistive control were evaluated during daily activities under both non-fatigued and fatigued conditions as shown in Fig. 8. A single participant performed four lower-limb tasks—walking, stair ascent, sitting down, and standing up—each repeated 10 times in a rested state and after a 10-minute jogging-induced fatigue protocol. Using 600-sample CMS windows, the system classified both motion type and fatigue state. As summarized in Table III, high recognition accuracy was achieved across all tasks, with stable performance maintained under fatigue, demonstrating robustness to dynamically changing muscle conditions.

In addition to accurate recognition, the system provided timely and adaptive assistive torque output. A preliminary assistive torque of 3 N m was applied within 20 ms after detecting a ± 3 unit deviation in CMS signal amplitude, helping the user initiate motion. Concurrently, a feature extraction and classification pipeline processed the corresponding 600-sample window. Based on the classification outcome, a refined torque output (typically 6 Nm) was applied at 62 ms to better match the user's motion intention and fatigue level. This two-stage torque control allowed the system to respond rapidly while maintaining adaptability.

The experimental results confirm that the proposed system is capable of maintaining high recognition accuracy and effective torque response across both normal and fatigued conditions. Its robustness to real-world factors such as muscle fatigue and signal variation supports its potential for long-term deployment in assistive wearable systems and prosthetic applications. To assess the effect of torque assistance under fatigue, five participants performed 15-min continuous tasks (sitting, standing,

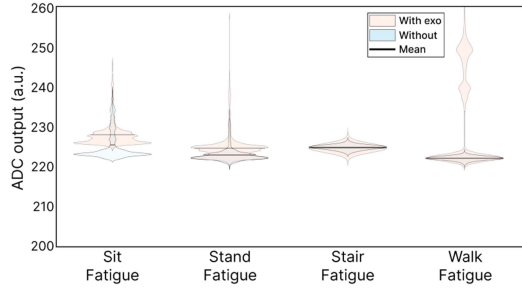


Fig. 9. MMG signal distributions after 15-min prolonged tasks under fatigued conditions, with and without exoskeleton assistance.

stair ascent, and walking) with and without the exoskeleton. MMG signals were recorded from the rectus femoris after task completion.

As shown in Fig. 9, the assisted condition exhibits broader and more evenly distributed MMG patterns across tasks, whereas the unassisted condition shows more concentrated distributions with higher peak densities after prolonged exertion. This redistribution suggests that the assistance torque alters the distribution of muscular effort under fatigue, consistent with reduced localized high-load activation.

Subjective feedback supported these observations: four out of five participants reported clearly perceivable assistance under fatigue, while the remaining participant reported weaker but noticeable assistance. All participants reported no discomfort or forced motion and described the interaction as comfortable.

C. Robustness Evaluation Under Realistic Conditions

All experiments were conducted under extended continuous usage conditions lasting approximately 1–2 hours, covering repeated dynamic movements relevant to wearable exoskeleton operation. Throughout these sessions, the CMS signals exhibited stable amplitude and temporal characteristics without noticeable degradation.

To assess robustness under realistic wearable conditions, CMS signal stability was evaluated under common perturbations, including attachment pressure variation, perspiration, loose contact, and environmental noise. Signal strength was quantified by RMS amplitude after DC removal and averaged across channels. A nominal skin–sensor load of 0.8–1.0 N was calibrated using an FSR402, with increased pressure set to 1.0–1.2 N. Perspiration was simulated by applying 5 ml saline, loose contact by lightly suspending the sensor, and noise robustness tested at 60–70 dB ambient sound levels. For each condition, 15 repeated stand-up trials were performed on the same participant.

$$\text{RMS}_j = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_{i,j} - \bar{x}_j)^2}, \quad \text{RMS}_{\text{avg}} = \frac{1}{C} \sum_{j=1}^C \text{RMS}_j$$

where $x_{i,j}$ is the i -th sample from channel j , $\bar{x}_j$ is the channel mean, $C = 5$ is the number of channels, and N is the number of samples.

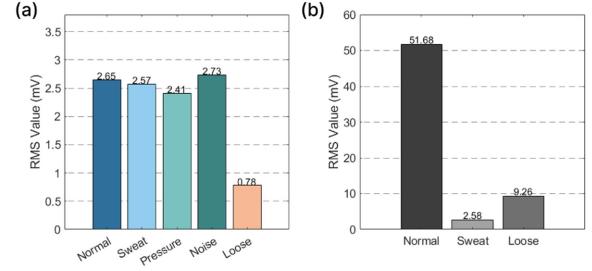


Fig. 10. (a) CMS RMS during the stand-up task remains stable across interference conditions, decreasing only under loose attachment. (b) EMG baseline RMS shows strong sensitivity to sweat and loose electrode–skin contact.

Results show minimal RMS variation under sweat (2.57 mV), pressure (2.41 mV), and noise (2.73 mV) compared to the baseline (2.65 mV), with changes within 10% as shown in Fig. 10. In contrast, the loose-wear condition caused a significant drop to 0.78 mV, indicating strong dependence on sensor–skin contact.

Robustness of the proposed CMS system was evaluated by analyzing RMS signal amplitude under common wearable perturbations, including attachment pressure variation, perspiration, loose contact, and environmental noise. RMS values were computed after DC offset removal and averaged across channels. Nominal attachment was calibrated using an FSR402 sensor to achieve an interface load of 0.8–1.0 N, while increased pressure (1.0–1.2 N), simulated perspiration (5 ml saline), loose contact, and ambient noise (60–70 dB) were applied. Each condition consisted of 15 repeated stand-up trials within the same participant.

To quantitatively characterize the magnitude of perturbation-induced changes beyond percentage-based descriptions, an effect-size analysis was conducted using a within-subject paired Cohen’s d_z . For each perturbation condition, RMS values from $n = 15$ repeated stand-up trials were compared against the baseline condition from the same participant. Trial-wise differences were computed as

$$d_z = \frac{\bar{\Delta}}{\text{SD}(\Delta)}, \quad \Delta_i = \text{RMS}_i^{(\text{cond})} - \text{RMS}_i^{(\text{base})}, \quad (4)$$

The resulting effect sizes reveal systematic RMS amplitude shifts under all tested perturbations. Sweat and ambient noise induced large effects ($d_z = -1.89$ and $d_z = +1.67$, respectively), while increased attachment pressure produced a very large negative effect ($d_z = -3.73$), reflecting enhanced mechanical coupling at the sensor–skin interface; in contrast, loose attachment caused a pronounced signal degradation with a huge negative effect size ($d_z = -5.36$). Despite these amplitude shifts under sweat, pressure, and noise, CMS signals remained continuous and usable for movement detection and control, without saturation or loss of discriminability, indicating *functional robustness* under realistic wearable perturbations, whereas loose attachment represents a clear failure mode requiring sufficient sensor–skin contact.

For comparison, EMG baseline measurements were conducted under normal, perspiration, and loose attachment conditions using the same experimental protocol. In practice, excessive perspiration naturally degraded electrode adhesion, leading

to partial detachment and behaviorally equivalent loose-contact conditions. This coupling between sweat and attachment instability resulted in a pronounced reduction in EMG signal quality, whereas CMS maintained more consistent RMS responses under the same conditions.

V. CONCLUSION

A CMS-based muscle sensing and control framework for wearable exoskeletons is presented, enabling early activity prediction and adaptive assistance through a two-stage torque strategy. Capacitive microphone sensors capture anticipatory muscle vibrations to trigger immediate pre-assist, followed by logistic-regression-based torque modulation for user-specific adaptation under motion variability and fatigue. Experiments across daily activities—including sitting, standing, stair ascent, and walking—demonstrate reliable recognition accuracy and timing in both non-fatigued and fatigued states. Fatigue-oriented evaluations indicate more balanced muscle activation and reduced localized overload with assistance, consistent with positive user feedback during prolonged use. Compared with EMG, CMS maintains stable signal characteristics under realistic wearable conditions, where perspiration and loose electrode contact degrade EMG reliability.

The integration of lightweight signal processing, robust CMS sensing, and machine-learning-based adaptation provides a scalable and user-friendly solution for real-world exoskeleton control. Future work will extend this framework with direct biomechanical and physiological assessments (e.g., EMG analysis and metabolic cost), multi-user studies, long-term evaluations, and embedded learning for continual performance improvement.

REFERENCES

- [1] A. Liu, Y. You, Y. Wang, L. Li, and J. Yuan, "Research on intrinsic capacity as a predictor of falls and disability in community-dwelling elderly," *Front. Aging*, vol. 6, 2025, Art. no. 1589369.
- [2] World Health Organization, "Ageing and Health," 2022. Accessed: Oct. 1, 2022.
- [3] P. Maresova et al., "Challenges and opportunity in mobility among older adults—Key determinant identification," *BMC Geriatrics*, vol. 23, no. 1, 2023, Art. no. 447.
- [4] S. Song and S. H. Collins, "Optimizing exoskeleton assistance for faster self-selected walking," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 786–795, 2021.
- [5] S. Luo et al., "Experiment-free exoskeleton assistance via learning in simulation," *Nature*, vol. 630, no. 8016, pp. 353–359, Jun. 2024.
- [6] J. Zhang et al., "Human-in-the-loop optimization of exoskeleton assistance during walking," *Science*, vol. 356, no. 6344, pp. 1280–1284, 2017.
- [7] A. B. Zoss, H. Kazerooni, and A. Chu, "Biomechanical design of the berkeley lower extremity exoskeleton (BLEEX)," *IEEE/ASME Trans. Mechatron.*, vol. 11, no. 2, pp. 128–138, Apr. 2006.
- [8] A. J. Young and D. P. Ferris, "State of the art and future directions for lower limb robotic exoskeletons," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 2, pp. 171–182, Feb. 2017.
- [9] G. S. Sawicki, O. N. Beck, I. Kang, and A. J. Young, "The exoskeleton expansion: Improving walking and running economy," *J. NeuroEng. Rehabil.*, vol. 17, no. 1, 2020, Art. no. 25.
- [10] O. Zeeshan Khokhar, Zhen G. Xiao, and C. Menon, "Surface EMG pattern recognition for real-time control of a wrist exoskeleton," *BioMedical Eng. OnLine*, vol. 9, no. 1, 41, 2010, Art. no. 41.
- [11] M. DiCicco, L. Lucas, and Y. Matsuoka, "Comparison of control strategies for an EMG controlled orthotic exoskeleton for the hand," in *Proc. IEEE Int. Conf. Robot. Automat.*, vol. 2, 2004, pp. 1622–1627.
- [12] K. Kiguchi, T. Tanaka, and T. Fukuda, "Neuro-fuzzy control of a robotic exoskeleton with EMG signals," *IEEE Trans. Fuzzy Syst.*, vol. 12, no. 4, pp. 481–490, Aug. 2004.
- [13] D. Leonardis et al., "An EMG-controlled robotic hand exoskeleton for bilateral rehabilitation," *IEEE Trans. Haptics*, 8, no. 2, pp. 140–151, 2015.
- [14] A. Císnal, J. Pérez-Turiel, J.-C. Fraile, D. Sierra, and E. de la Fuente, "Rob-Hand: A hand exoskeleton with real-time EMG-driven embedded control. quantifying hand gesture recognition delays for bilateral rehabilitation," *IEEE Access*, 9, pp. 137809–137823, 2021.
- [15] B. Treussart, F. Geffard, N. Vignais, and F. Marin, "Controlling an upper-limb exoskeleton by EMG signal while carrying unknown load," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 9107–9113.
- [16] S. Y. Gordleeva et al., "Real-time EEG-EMG human-machine interface-based control system for a lower-limb exoskeleton," *IEEE Access*, 8, pp. 84070–84081, 2020.
- [17] G. Yin et al., "Processing surface EMG signals for exoskeleton motion control," *Front. Neurobot.*, vol. 14, 2020, Art. no. 40.
- [18] C. Sebastian Mancero Castillo, S. Wilson, R. Vaidyanathan, and S. F. Atashzar, "Wearable MMG-plus-one armband: Evaluation of normal force on mechanomyography (MMG) to enhance human-machine interfacing," *IEEE Trans. Neural Syst. Rehabil. Eng.*, 29, pp. 196–205, 2021.
- [19] C. Caulcrick, W. Huo, W. Hoult, and R. Vaidyanathan, "Human joint torque modelling with MMG and EMG during lower limb human-exoskeleton interaction," *IEEE Robot. Automat. Lett.*, 6, no. 4, pp. 7185–7192, Oct. 2021.
- [20] N. Hondo and T. Tsuji, "Knee joint torque estimation in femoral prosthesis users using MMG and low-frequency acceleration," *IEEE Access*, vol. 12, pp. 174996–175005, 2024.
- [21] A. Guez, J. Li, Y. Chen, Y. Wang, and R. Vaidyanathan, "MMG-based real-time wrist tremor detection for exoskeleton integration," in *Proc. IEEE Int. Conf. Cyborg Bionic Syst.*, 2024, pp. 154–159.
- [22] X. Jin, B. Xiao, H. Wang, W. Wang, P. Childs, and Z. Yu, "A multi-sensor wearable device based on capacitive microphone sensors for robust human-robot interaction in smart home environments," *IEEE Sensors J.*, vol. 26, no. 2, pp. 2935–2945, Jan. 2026.
- [23] S. Yu et al., "Controlling negative and positive power for efficiency enhancement and muscle strain mitigation during squatting with a portable knee exoskeleton," *Ann. Biomed. Eng.*, vol. 53, pp. 1344–1358, 2025.
- [24] A. Arefeen and Y. Xiang, "Subject specific optimal control of powered knee exoskeleton to assist human lifting tasks under controlled environment," *Robotica*, vol. 41, pp. 2809–2828, 2023.